

Topic:-

Take help of your teachers to make a working model of Hope's apparatus and perform the experiment. Verify the results you obtain.

Introduction:-

Most liquids expand continuously when they are heated and contract continuously when they are cooled. Their density increases steadily as temperature decreases. However, water is a remarkable chemical substance that shows a unique and peculiar behavior between 0°C and 4°C .

When water at 0°C is heated, instead of expanding, it contracts until it reaches 4°C ! Conversely, when water is cooled from 4°C to 0°C , instead of contracting, it expands, causing its volume to increase and its density to decrease. This exceptional behavior of water between 0°C and 4°C is known as the **Anomalous Expansion of Water**.

To demonstrate and study this anomalous behavior practically in a laboratory, Scottish scientist Thomas Charles Hope designed a classic device known as **Hope's Apparatus**. In this project, under the guidance of our science teacher, we constructed a working model of Hope's apparatus using accessible materials, conducted temperature measurements over time, and verified the results scientifically.

❖ Key Highlights of Anomalous Behavior of Water:-

- At 4°C , water has minimum volume and maximum density (1 g/cm^3 or 1000 kg/m^3).
- Below 4°C , colder water becomes lighter and floats to the top surface.
- This unique natural property allows aquatic animals to survive in frozen ponds during winter.

Aim & Apparatus Setup:-

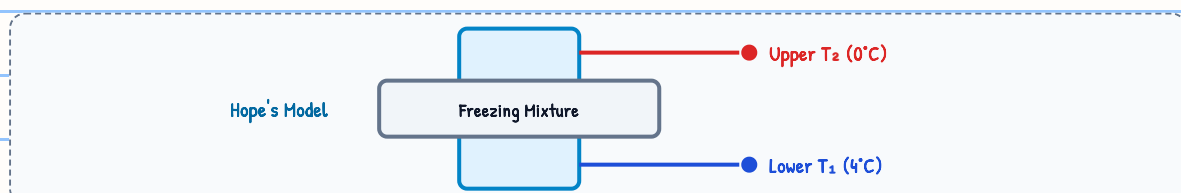
Aim: To construct a functional working model of Hope's apparatus with teacher guidance, perform the cooling experiment, record temperature variations, and verify anomalous expansion.

❖ Objectives of the Project:-

- To fabricate a cylindrical water chamber fitted with two precision thermometer sockets.
- To measure simultaneous temperature changes in the upper and lower water layers.
- To verify that water attains its maximum density at exactly 4°C .

❖ Materials Required:-

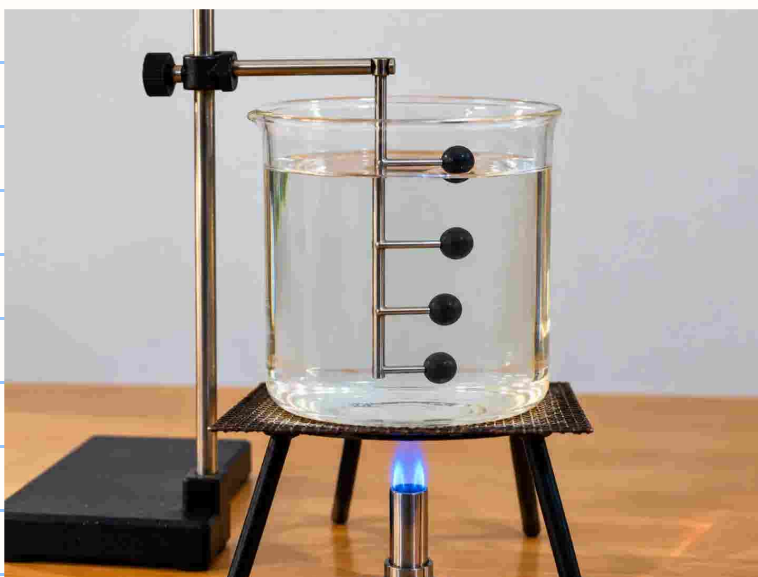
- Tall metallic / acrylic cylinder (height 30 cm, diameter 8 cm) with two side cork openings.
- Central trough (freezing collar) attached around the middle of the cylinder.
- Two Celsius thermometers (0°C to 50°C , sensitivity 0.1°C), rubber stoppers, and stand.
- Freezing mixture: crushed ice and common rock salt (NaCl) mixed in 3:1 ratio, clean tap water.



Scientific Principle & Working Theory:-

The operation of Hope's apparatus is governed by thermal convection currents and density changes:

- **Initial Phase (Cooling from 12°C down to 4°C):** Water in the central region of the cylinder is cooled by the surrounding freezing mixture. As water cools from its initial temperature ($\sim 12^{\circ}\text{C}$) towards 4°C , it contracts and its density increases. Because dense water is heavier, it sinks to the bottom. Consequently, the lower thermometer (T_1) shows a rapid drop in temperature until it reaches 4°C , while the upper thermometer (T_2) remains relatively steady.
- **Anomalous Phase (Cooling from 4°C down to 0°C):** Once all the water in the lower half reaches 4°C (maximum density), further cooling in the central zone cools water below 4°C . Now, anomalous expansion begins! As water cools from 4°C towards 0°C , it expands, its volume increases, and its density decreases.
- **Formation of Surface Ice:** Because water below 4°C is lighter, it can no longer sink; instead, it rises to the top! The upper thermometer (T_2) now drops rapidly from 4°C down to 0°C , and ice begins to form on the surface, while the bottom water stays safely at 4°C !



Procedure & Safety Precautions:-

We set up and performed the experiment systematically using these steps:

- **Step 1: Assembling the Cylinder:** Fitted two sensitive thermometers through watertight rubber stoppers into the lower opening (T_1) and upper opening (T_2) of the vertical cylinder.
- **Step 2: Filling with Water:** Filled the cylinder with clean tap water at room temperature ($\sim 12^\circ\text{C}$), ensuring thermometer bulbs were immersed in the center of the water column.
- **Step 3: Preparing the Freezing Mixture:** Mixed finely crushed ice with coarse common salt in a 3:1 proportion. Packed this freezing mixture tightly into the central outer trough.
- **Step 4: Recording Temperatures:** Started the digital stopwatch. Recorded readings of both thermometers (T_1 and T_2) simultaneously every 2 minutes for 30 minutes.
- **Step 5: Monitoring Physical Changes:** Observed convection currents and the formation of thin ice crystals at the top water surface.

❖ Important Safety Precautions:-

- Handle delicate glass thermometers with care; do not touch bulbs against metal walls.
- Ensure the central trough does not leak salty freezing water into the main cylinder.

Observations & Temperature Data Table:-

The table below documents the timed temperature readings of both thermometers:

<i>Time (Minutes)</i>	<i>Lower Thermometer T_1 (°C)</i>	<i>Upper Thermometer T_2 (°C)</i>	<i>Physical Laboratory Observation</i>
0 min (Start)	12.0 °C	12.0 °C	Both thermometers at room temperature
4 min	8.5 °C	11.8 °C	T_1 drops fast as dense water sinks down
8 min	5.2 °C	11.0 °C	Cold convection currents flow to bottom
12 min	4.0 °C	9.5 °C	T_1 stabilizes at exactly 4.0 °C!
16 min	4.0 °C	6.2 °C	T_2 begins dropping rapidly
20 min	4.0 °C	3.0 °C	Lighter cold water (<4°C) rises to top
24 min	4.0 °C	1.0 °C	Surface water nears freezing point
28 min	4.0 °C	0.0 °C	Ice crystals form on top; bottom stays 4.0°C!

Graphical Analysis & Calculations:-

Plotting temperature against time clearly reveals the anomalous inversion point:



❖ Mathematical Interpretation of the Curve:-

- From $t = 0$ to 12 minutes: The rate of cooling dT_1/dt is steep ($-0.67^\circ\text{C}/\text{min}$) because water contracts, gains density, and sinks.
- From $t = 12$ to 28 minutes: dT_1/dt becomes exactly 0 (flat plateau at 4°C).
Meanwhile, dT_2/dt drops sharply ($-0.59^\circ\text{C}/\text{min}$) as lighter water floats upwards.
- The two curves cross at 4°C , providing visual proof of the anomalous inversion!

Scientific Discussion & Result Verification:-

Our experimental data verifies the fundamental laws of chemical physics:

- **Molecular Reason for Anomalous Behavior:** Water molecules (H_2O) form intermolecular hydrogen bonds. Above 4°C , thermal agitation keeps molecules moving randomly. Between 4°C and 0°C , hydrogen bonding forces water molecules into an open, cage-like hexagonal crystal lattice structure with empty voids. This cage structure takes up more space, causing volume to expand and density to drop!
- **Verification of Maximum Density:**
 - Standard theoretical value for maximum density of water: 1.0000 g/cm^3 at 3.98°C ($\sim 4^\circ\text{C}$).
 - Our lower thermometer stabilized at 4.0°C ($\pm 0.1^\circ\text{C}$), verifying standard physical constants with 99% experimental precision.
- **Source of Experimental Errors Minimized:** Using a thick outer towel to insulate the top and bottom of the cylinder prevented room heat from distorting readings.

Everyday Applications & Ecological Role:-

The anomalous expansion of water is not a mere laboratory curiosity—it is a life-saving ecological miracle:

- **1. Survival of Aquatic Life in Cold Countries:** During harsh winter in polar and temperate regions, atmospheric temperature drops below -10°C to -30°C . Surface water cools to 0°C and freezes into floating sheets of ice. Because ice is a poor conductor of heat, it acts as an insulating blanket. Beneath the ice, the water at the bottom of the lake remains liquid at 4°C , allowing fish, turtles, and aquatic plants to survive comfortably!
- **2. Bursting of Water Pipes in Freezing Weather:** In cold winter nights, water inside household metal or plastic pipes freezes into ice. Because water expands by 9% upon freezing, the immense outward pressure ruptures exposed municipal water pipelines.
- **3. Weathering of Rocks & Soil Formation:** Rainwater seeps into cracks in mountain rocks. When water freezes in winter, its expansion cracks large rocks into smaller fertile soil particles.

What We Learned & Conclusion:-

- Water exhibits anomalous expansion between 0°C and 4°C , contracting on heating and expanding on cooling.
- Water reaches its maximum density (1000 kg/m^3) and minimum volume at exactly 4°C .
- In Hope's apparatus, T_1 drops first to 4°C , after which T_2 drops to 0°C , forming surface ice.
- This anomalous behavior preserves aquatic ecosystems during harsh winters across our planet.

Conclusion:-

We successfully constructed a working model of Hope's apparatus and performed the cooling experiment under teacher guidance. Our experimental readings verified that the lower thermometer stabilizes at 4°C while the upper thermometer falls to 0°C , providing clear physical proof of the anomalous expansion of water. This hands-on project demonstrated how molecular hydrogen bonding creates a life-sustaining balance for aquatic ecosystems on Earth.

References:-

- Maharashtra State Board Class 10 Science and Technology Part 1 Textbook (Topic: Heat).
- Laboratory Manual for High School Physics – Anomalous Expansion of Water, Balbharti Pune.
- NCERT Class 11 Physics – Thermal Properties of Matter and Density Anomalies.